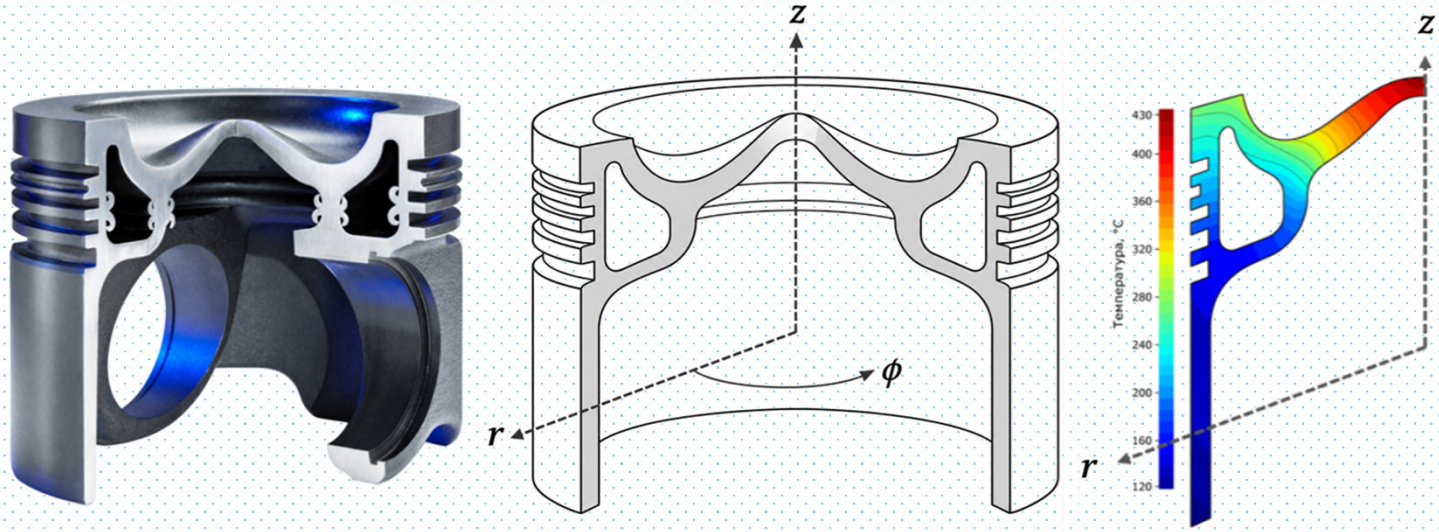
Bauman Moscow State Technical University

Special Topics in Finite Element Modeling for IC engines

Finite Element Analysis of the Steady-State Thermal Field of a Piston in an Axisymmetric Formulation

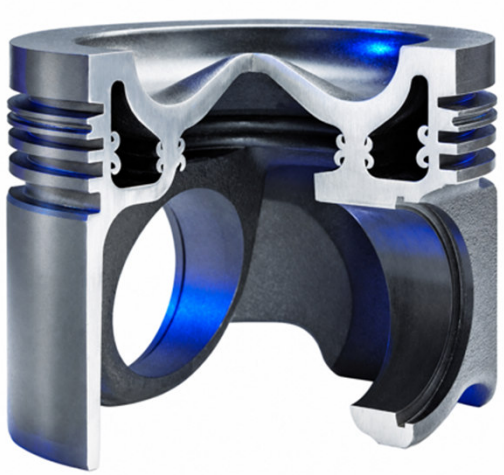


Teachers: Malastowski Nikolay, Blinov Artem

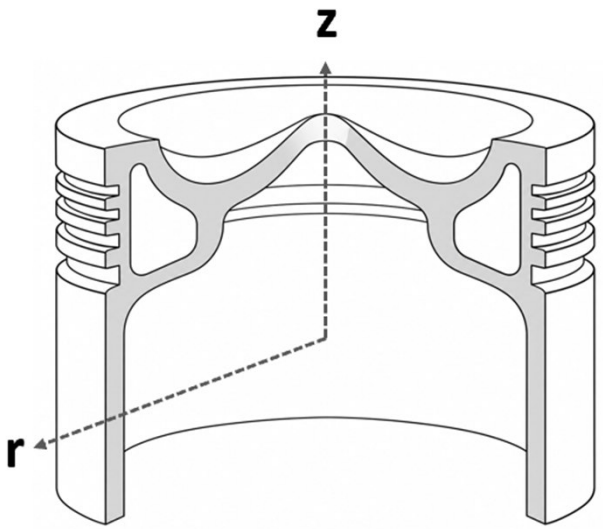
Problem statement

Steady-state temperature field of a piston

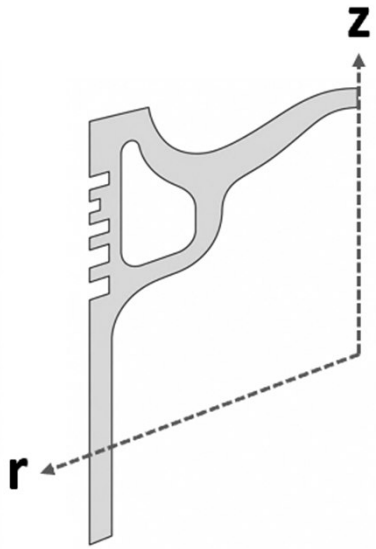
Axisymmetric model in a meridional cross-section



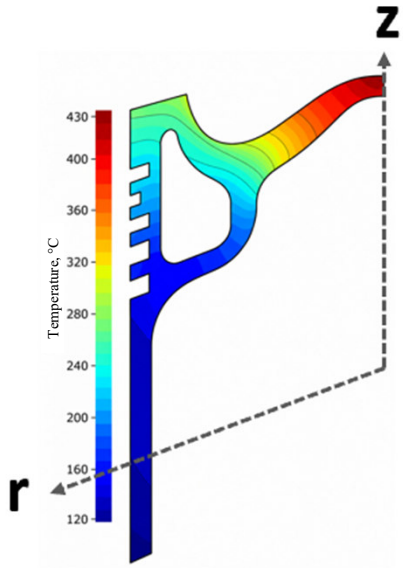
Component geometry



Axisymmetric model



Meridional cross-section



Temperature field

Goal: determine T = T(r, z) for the prescribed boundary conditions.

Heat balance equation

Three-dimensional heat conduction equation

1. Heat balance

$$\int_V \rho c \frac{\partial T}{\partial t}\, dV = -\int_{\partial V} \mathbf{q}\cdot\mathbf{n}\, dS + \int_V Q\, dV$$

Heat accumulation = boundary heat input
+ internal heat generation.

2. Local governing equation

$$\rho c \frac{\partial T}{\partial t} = -\nabla\cdot\mathbf{q} + Q$$

The divergence theorem and arbitrary volume V give the balance at every point.

3. Substitution of Fourier's law

$$\mathbf{q} = -\boldsymbol{\Lambda}\nabla T$$

$$\rho c \frac{\partial T}{\partial t} = \nabla\cdot(\boldsymbol{\Lambda}\nabla T) + Q$$

Λ: thermal conductivity tensor.

4. Assumption of isotropy

$$\boldsymbol{\Lambda} = \lambda\mathbf{I}$$

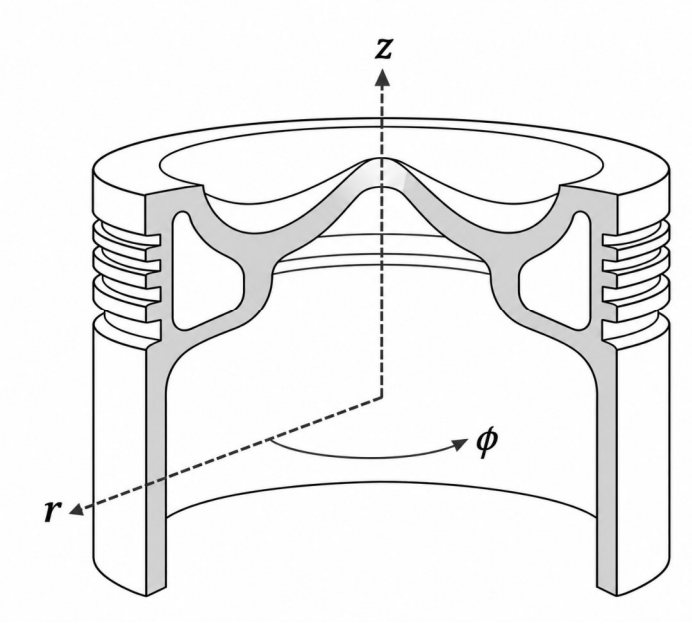
$$\rho c \frac{\partial T}{\partial t} = \nabla\cdot(\lambda\nabla T) + Q$$

λ is the same in every direction.

Isotropy does not imply λ = const. Specify the initial field and boundary conditions separately.

Axisymmetric formulation

Cylindrical coordinates and heat conduction



1. Coordinate definitions

r: distance from the axis of rotation;
φ: angle of rotation about the z-axis;
z: coordinate along the axis.

The material is isotropic.
Thermal conductivity λ may depend on position and temperature.

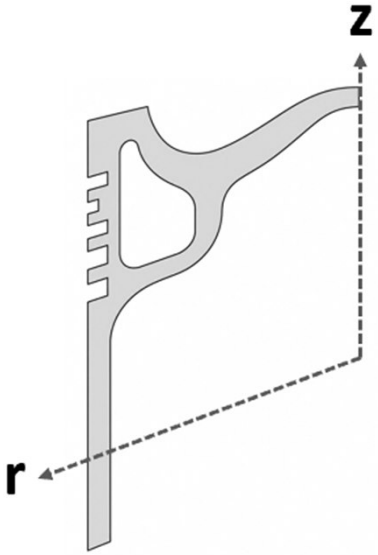
2. General equation in cylindrical coordinates

$$\rho c \frac{\partial T}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(r\lambda\frac{\partial T}{\partial r}\right) + \frac{1}{r^2}\frac{\partial}{\partial \varphi}\left(\lambda\frac{\partial T}{\partial \varphi}\right) + \frac{\partial}{\partial z}\left(\lambda\frac{\partial T}{\partial z}\right) + Q$$

All three coordinates, time dependence, and internal heat sources remain.

Axisymmetric formulation

Axisymmetric temperature field model



1. Rotationally invariant conditions

Geometry, properties, heat sources, boundary conditions, and initial conditions do not depend on φ.

2. Temperature is independent of φ

$$T = T(r, z, t), \qquad \frac{\partial T}{\partial \varphi} = 0$$

3. The angular term vanishes

$$\rho c \frac{\partial T}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(r\lambda\frac{\partial T}{\partial r}\right) + \frac{\partial}{\partial z}\left(\lambda\frac{\partial T}{\partial z}\right) + Q$$

Axisymmetry does not imply steady state: temperature depends on time and Q remains.

Steady-state model with constant thermal conductivity

1. Model assumptions

Steady state and a homogeneous, isotropic material.

$$\frac{\partial T}{\partial t} = 0, \qquad \mathbf{\Lambda} = \lambda \mathbf{I}, \qquad \lambda = \text{const} > 0$$

Temperature T = T(r, z). The source Q remains.

2. Heat balance

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\lambda\frac{\partial T}{\partial r}\right) + \frac{\partial}{\partial z}\left(\lambda\frac{\partial T}{\partial z}\right) + Q = 0$$

There is no heat accumulation over time.

3. Expansion of the derivatives

Take constant λ outside the derivatives, divide by λ, and expand the radial derivative.

$$\frac{\partial^2 T}{\partial r^2} + \frac{1}{r}\frac{\partial T}{\partial r} + \frac{\partial^2 T}{\partial z^2} + \frac{Q}{\lambda} = 0, \qquad r > 0$$

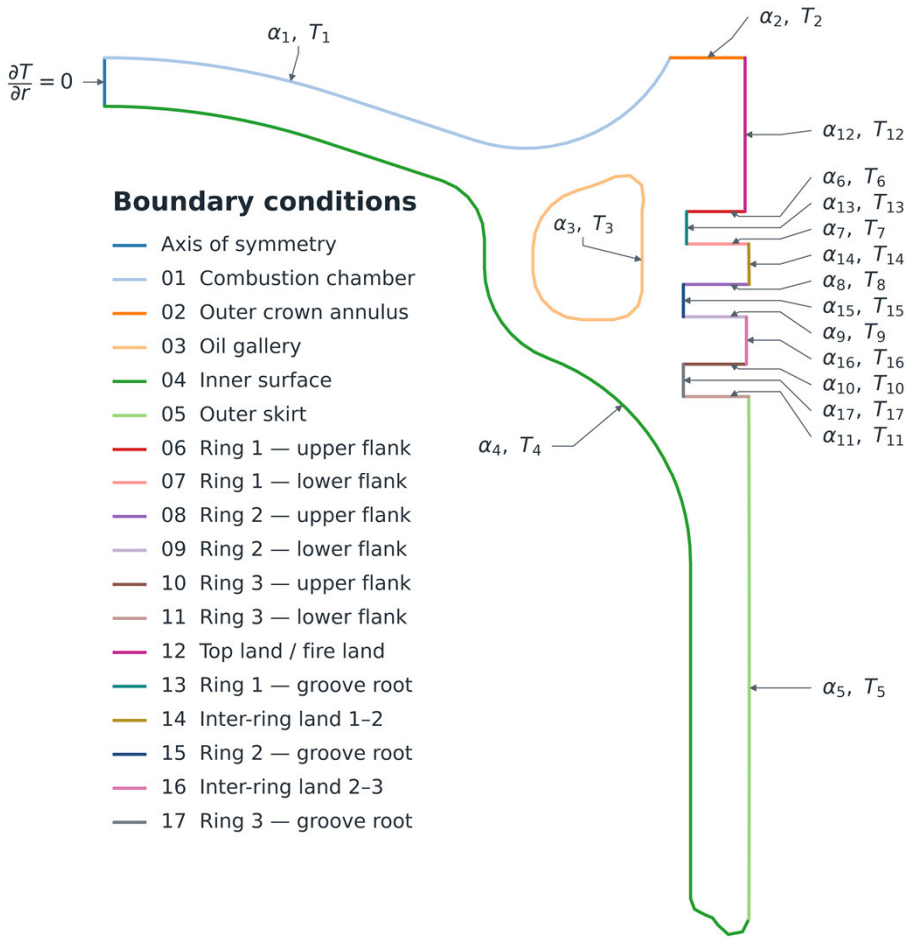
4. Axisymmetric geometry

$$\frac{1}{r}\frac{\partial T}{\partial r}, \qquad dS_r = 2\pi r\, dz$$

The cylindrical heat-transfer surface area varies with the radius.

The term (1/r) ∂T/∂r accounts for axisymmetric geometry, not heat generation or circumferential transfer.

Thermal boundary conditions



The normal n points out of the body

$$q_n = \mathbf{q}\cdot\mathbf{n} = -\lambda\frac{\partial T}{\partial n}$$

Positive heat flux represents heat loss.

Dirichlet: prescribed temperature

$$T = \overline{T}$$

The surface temperature is known.

Neumann: prescribed heat flux

$$-\lambda\frac{\partial T}{\partial n} = \overline{q}_n$$

For $\bar{q}_n = 0$, the boundary is insulated.

Robin: convection to the surroundings

$$-\lambda\frac{\partial T}{\partial n} = \alpha(T - T_\infty)$$

Prescribe $\alpha \geq 0$ and $T\infty$. Solve for T.

Robin conditions apply on heat-transfer surfaces of the piston; the surface temperature is unknown.
On the axis, symmetry requires $\partial T/\partial r = 0$.

Variational approach to the problem

1. A number assigned to each field

A functional assigns a number
to a temperature field.

$$T \longmapsto \Phi[T]$$

Consider admissible temperature fields.

2. The field that minimizes the functional

$$\Phi[T^*] = \min_{T \text{ admissible}} \Phi[T]$$

T*: the sought field. Minimize the functional,
not the temperature itself.

3. Preservation of the original heat balance

For our linear steady-state problem, choose Φ
so that its minimum condition yields
the original heat conduction equation
and the prescribed boundary conditions.

$$\delta\Phi = 0 \iff \begin{array}{c}\text{original heat equation} \\ + \text{ prescribed boundary conditions}\end{array}$$

Both approaches give the same field T*.

4. FEM: transition to nodal temperatures

$$\Phi[T] \longrightarrow \Phi_h(T_1, \ldots, T_N)$$

$$\frac{\partial \Phi_h}{\partial T_i} = 0, \quad i = 1, \ldots, N \quad \Longrightarrow \quad \mathbf{KT} = \mathbf{F}$$

Solve the system and recover the field.

Finite element discretization produces the system of algebraic equations.

Heat conduction functional

General functional: four physical contributions

1. Functional over the volume V

Linear steady-state problem: $\lambda > 0$ is prescribed and independent of the unknown T.

$$\Phi[T] = \int_V \left[\frac{\lambda}{2}|\nabla T|^2 - QT\right] dV + \int_{S_2} \bar{q}_n T \, dS + \frac{1}{2}\int_{S_3} \alpha (T - T_\infty)^2 \, dS.$$

2. Meaning of the terms

λ: thermal conductivity of the material;
Q: volumetric heat source.

$\bar{q}_n$: prescribed outward heat flux on S_2;
α: heat transfer coefficient on S_3.

Dirichlet condition: $T = \bar{T}$ on S_1 constrains the admissible temperature field.

The positive sign of the prescribed-flux contribution corresponds to outward heat flux.

Axisymmetric functional

1. Rotation of the meridional cross-section

ω: cross-section; Γ_2, Γ_3: lines that generate the surfaces S_2, S_3.

$$dA = dr\,dz, \qquad dV = 2\pi r\,dA, \qquad dS = 2\pi r\,dL$$

The factor 2πr is the circumference about the z-axis.

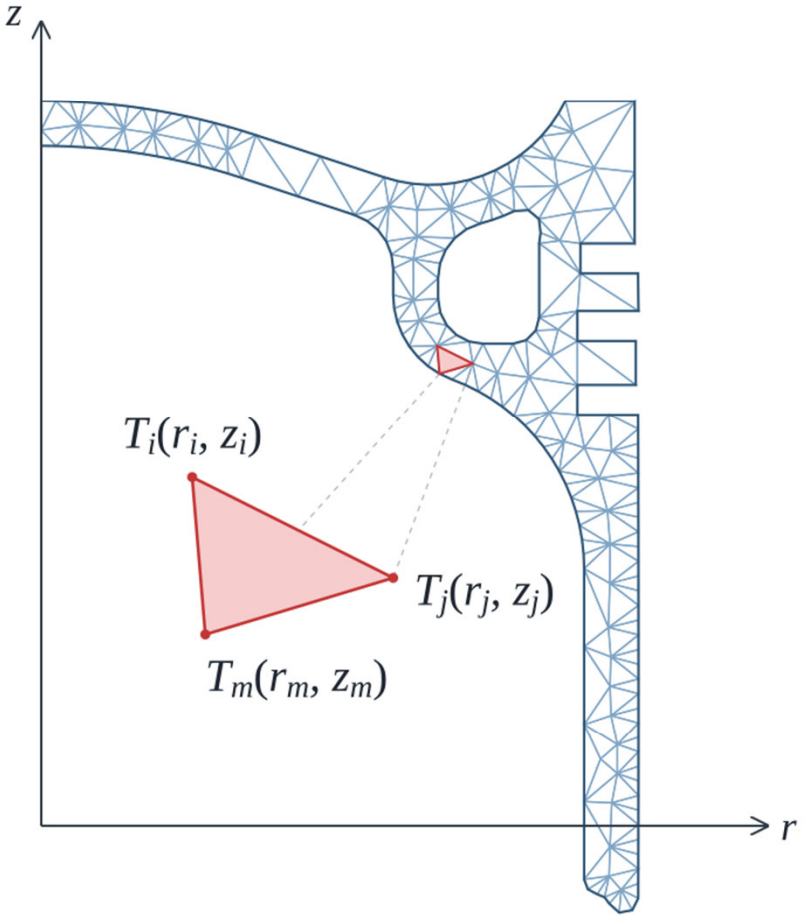
2. All terms expressed in coordinates r, z

$$\Phi[T] = 2\pi \int_{\omega} \left[\frac{\lambda}{2} \left(\left(\frac{\partial T}{\partial r} \right)^2 + \left(\frac{\partial T}{\partial z} \right)^2 \right) - QT \right] r\,dA + 2\pi \int_{\Gamma_2} \overline{q}_n T r\,dL + \pi \int_{\Gamma_3} \alpha (T - T_\infty)^2 r\,dL.$$

The weight 2πr accounts for axisymmetric geometry in both volume and surface integrals.

Temperature approximation from nodal values

Linear approximation in a triangle



1. Polynomial temperature approximation within the finite element

In element e with vertices i, j, m:

$$T^e(r, z) = \beta_0 + \beta_r r + \beta_z z$$

2. Determination of three coefficients

From the temperatures at the three nodes:

$$\underbrace{\begin{pmatrix} 1 & r_i & z_i \\ 1 & r_j & z_j \\ 1 & r_m & z_m \end{pmatrix}}_{\mathbf{X}_e} \begin{pmatrix} \beta_0 \\ \beta_r \\ \beta_z \end{pmatrix} = \underbrace{\begin{pmatrix} T_i \\ T_j \\ T_m \end{pmatrix}}_{\mathbf{T}_e}$$

3. Nondegeneracy check

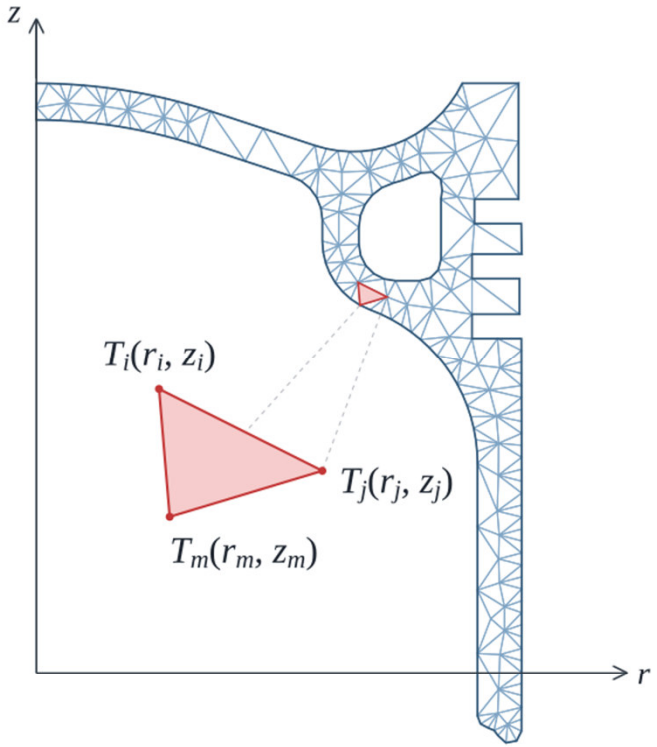
$$D_e = \det \mathbf{X}_e \neq 0, \qquad A_e = \frac{|D_e|}{2}$$

D_e: signed determinant; A_e: area.

Three nodal temperatures fully determine the linear temperature field in a triangle.

Temperature approximation from nodal values

From polynomial coefficients to shape functions



1. Solution for the polynomial coefficients

$$\beta_0 = \frac{\sum_a a_a T_a}{D_e} = \frac{a_i T_i + a_j T_j + a_m T_m}{D_e},$$
$$\beta_r = \frac{\sum_a b_a T_a}{D_e} = \frac{b_i T_i + b_j T_j + b_m T_m}{D_e},$$
$$\beta_z = \frac{\sum_a c_a T_a}{D_e} = \frac{c_i T_i + c_j T_j + c_m T_m}{D_e}.$$

Index a runs over nodes i, j, m.

2. Coefficients depend on vertex coordinates

$$(a_i, a_j, a_m) = (r_j z_m - r_m z_j,\ r_m z_i - r_i z_m,\ r_i z_j - r_j z_i),$$
$$(b_i, b_j, b_m) = (z_j - z_m,\ z_m - z_i,\ z_i - z_j),$$
$$(c_i, c_j, c_m) = (r_m - r_j,\ r_i - r_m,\ r_j - r_i).$$

3. Substitution and grouping by T_a

$$T^e(r,z) = \frac{a_i + b_i r + c_i z}{D_e} T_i + \frac{a_j + b_j r + c_j z}{D_e} T_j + \frac{a_m + b_m r + c_m z}{D_e} T_m$$

$$T^e(r,z) = \sum_{a \in \{i,j,m\}} N_a(r,z) T_a, \quad N_a(r,z) = \frac{a_a + b_a r + c_a z}{D_e}$$ shape functions

Shape functions relate the temperature inside the element to its nodal temperatures.

Element functional in terms of nodal temperatures

1. Substitution of temperature interpolation

$$T^e = N_i T_i + N_j T_j + N_m T_m$$

2. Complete element functional

$$\Phi_e(T_i, T_j, T_m) = \pi \int_{A_e} \lambda r \left[\left(\sum_a \frac{\partial N_a}{\partial r} T_a \right)^2 + \left(\sum_a \frac{\partial N_a}{\partial z} T_a \right)^2 \right] dA - 2\pi \int_{A_e} Q r \left(N_i T_i + N_j T_j + N_m T_m \right) dA$$
$$+ 2\pi \int_{\Gamma_{2e}} \bar{q}_n r \left(N_i T_i + N_j T_j + N_m T_m \right) dL + \pi \int_{\Gamma_{3e}} \alpha r (N_i T_i + N_j T_j + N_m T_m - T_\infty)^2 dL.$$

Four physical contributions:

$$\Phi_e = \Phi_e^{(\lambda)} + \Phi_e^{(Q)} + \Phi_e^{(q)} + \Phi_e^{(\alpha)}$$

Expansion of derivatives with respect to r and z:

$$\sum_a \frac{\partial N_a}{\partial r} T_a = \frac{\partial N_i}{\partial r} T_i + \frac{\partial N_j}{\partial r} T_j + \frac{\partial N_m}{\partial r} T_m \qquad \sum_a \frac{\partial N_a}{\partial z} T_a = \frac{\partial N_i}{\partial z} T_i + \frac{\partial N_j}{\partial z} T_j + \frac{\partial N_m}{\partial z} T_m$$

Γ_{2e} and Γ_{3e} are parts of the physical boundary (external or internal).
Shared interior mesh edges do not enter these boundary integrals.

Shape function derivatives and the matrix B_e

1. Shape function derivatives with respect to coordinates

The geometric coefficients a_a, b_a, c_a and D_e are constant within the element.

$$N_a = \frac{a_a + b_a r + c_a z}{D_e} \qquad \frac{\partial N_a}{\partial r} = \frac{0 + b_a + 0}{D_e} = \frac{b_a}{D_e}, \qquad \frac{\partial N_a}{\partial z} = \frac{0 + 0 + c_a}{D_e} = \frac{c_a}{D_e}$$

2. Derivatives of temperature interpolation

The nodal temperatures T_i, T_j, T_m do not depend on coordinates r and z.

$$\frac{\partial T^e}{\partial r} = \frac{\partial N_i}{\partial r} T_i + \frac{\partial N_j}{\partial r} T_j + \frac{\partial N_m}{\partial r} T_m = \frac{b_i T_i + b_j T_j + b_m T_m}{D_e}$$

$$\frac{\partial T^e}{\partial z} = \frac{\partial N_i}{\partial z} T_i + \frac{\partial N_j}{\partial z} T_j + \frac{\partial N_m}{\partial z} T_m = \frac{c_i T_i + c_j T_j + c_m T_m}{D_e}$$

3. Matrix form of the derivatives

$$\begin{pmatrix} \frac{\partial T^e}{\partial r} \\ \frac{\partial T^e}{\partial z} \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{\partial N_i}{\partial r} & \frac{\partial N_j}{\partial r} & \frac{\partial N_m}{\partial r} \\ \frac{\partial N_i}{\partial z} & \frac{\partial N_j}{\partial z} & \frac{\partial N_m}{\partial z} \end{pmatrix}}_{\mathbf{B}_e} \begin{pmatrix} T_i \\ T_j \\ T_m \end{pmatrix}$$

$$\mathbf{B}_e = \frac{1}{D_e} \begin{pmatrix} b_i & b_j & b_m \\ c_i & c_j & c_m \end{pmatrix} = \frac{1}{D_e} \mathbf{H}_e$$

$$\nabla T^e = \mathbf{B}_e \mathbf{T}_e$$

The 2 × 3 matrix B_e maps three nodal temperatures to two derivatives.
The temperature gradient is constant within a linear triangular element.

Functional derivation: heat conduction

1. Original term

Constant thermal conductivity: $\lambda = \text{const} > 0$.

$$\Phi_e^{(\lambda)} = \pi\lambda \int_{A_e} r \left[\left(\frac{\partial T^e}{\partial r} \right)^2 + \left(\frac{\partial T^e}{\partial z} \right)^2 \right] dA$$

2. Substitution of temperature derivatives

$$\Phi_e^{(\lambda)} = \frac{\pi\lambda}{D_e^2} \int_{A_e} r \big[(b_i T_i + b_j T_j + b_m T_m)^2 + (c_i T_i + c_j T_j + c_m T_m)^2 \big] dA$$

3. Constants taken outside the integral

T_a, b_a, c_a do not depend on r or z; a = i, j, m.

$$\Phi_e^{(\lambda)} = \frac{\pi\lambda}{D_e^2} \left[\left(\sum_a b_a T_a \right)^2 + \left(\sum_a c_a T_a \right)^2 \right] \int_{A_e} r \, dA$$

4. Integration of the radial coordinate

$$\int_{A_e} r \, dA = A_e \bar{r}_e, \qquad \bar{r}_e = \frac{r_i + r_j + r_m}{3}$$

$$\Phi_e^{(\lambda)} = \frac{\pi\lambda A_e \bar{r}_e}{D_e^2} \left[\left(\sum_a b_a T_a \right)^2 + \left(\sum_a c_a T_a \right)^2 \right]$$

5. Expansion of the squares: a polynomial in nodal temperatures

$$\Phi_e^{(\lambda)} = \frac{\pi\lambda A_e \bar{r}_e}{D_e^2} \big[(b_i^2 + c_i^2) T_i^2 + (b_j^2 + c_j^2) T_j^2 + (b_m^2 + c_m^2) T_m^2 + 2(b_i b_j + c_i c_j) T_i T_j + 2(b_i b_m + c_i c_m) T_i T_m + 2(b_j b_m + c_j c_m) T_j T_m \big].$$

The spatial integral gives a scalar expression in T_i, T_j, T_m.
Nodal-temperature derivatives are taken later, during minimization.

Functional derivation: volumetric heat source

1. Original term

The source is constant within the element: Q = const.

$$\Phi_e^{(Q)} = -2\pi \int_{A_e} Q r T^e \, dA$$

2. Substitution of the temperature

$$\begin{aligned}\Phi_e^{(Q)} &= -2\pi Q \int_{A_e} r(N_i T_i + N_j T_j + N_m T_m)\, dA \\ &= -2\pi Q \left[T_i \int_{A_e} r N_i \, dA + T_j \int_{A_e} r N_j \, dA + T_m \int_{A_e} r N_m \, dA \right].\end{aligned}$$

3. Interpolation of the radial coordinate

$$r = r_i N_i + r_j N_j + r_m N_m$$

4. Exact integration

For a linear triangular element:

$$\int_{A_e} N_i^2 \, dA = \frac{A_e}{6} \qquad \int_{A_e} N_i N_j \, dA = \frac{A_e}{12}$$

In the second integral, i ≠ j; A_e is the element area.

5. Integrals weighted by the radius: node i

$$\int_{A_e} r N_i \, dA = r_i \int_{A_e} N_i^2 \, dA + r_j \int_{A_e} N_i N_j \, dA + r_m \int_{A_e} N_i N_m \, dA = \frac{A_e}{12}(2r_i + r_j + r_m)$$

6. Corresponding results for j and m and the final term

$$\Phi_e^{(Q)} = -\frac{\pi Q A_e}{6}\left[(2r_i + r_j + r_m)T_i + (r_i + 2r_j + r_m)T_j + (r_i + r_j + 2r_m)T_m\right]$$

The result is linear in the nodal temperatures.
The negative sign of the original functional is retained.

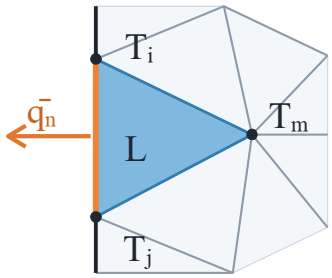
Functional derivation: prescribed heat flux

1. Original boundary term

Prescribed heat flux is positive outward.

$$\Phi_{\Gamma,e}^{(q)} = 2\pi \int_{\Gamma_e} \bar{q}_n \, r \, T^e \, dL$$

2. Temperature on the straight edge i, j



Domain boundary

L: edge length; $0 \le \xi \le 1$.

$$N_i = 1 - \xi, \quad N_j = \xi, \quad N_m = 0$$

$$T^e = N_i T_i + N_j T_j$$

$$r = r_i(1-\xi) + r_j \xi$$

$$dL = L \, d\xi$$

3. Constants taken outside the integral

Assume $\bar{q}_n$ = const on the edge.

$$\Phi_{\Gamma,e}^{(q)} = 2\pi \bar{q}_n \left[T_i \underbrace{\int_{\Gamma_e} r N_i \, dL}_{J_i} + T_j \underbrace{\int_{\Gamma_e} r N_j \, dL}_{J_j} \right]$$

4. Integration along the edge

$$J_i = L \int_0^1 [r_i(1-\xi) + r_j \xi](1-\xi) \, d\xi$$

$$= L \left(\frac{r_i}{3} + \frac{r_j}{6} \right) = \frac{L}{6}(2r_i + r_j),$$

$$J_j = L \int_0^1 [r_i(1-\xi) + r_j \xi] \xi \, d\xi = \tfrac{L}{6}(r_i + 2r_j).$$

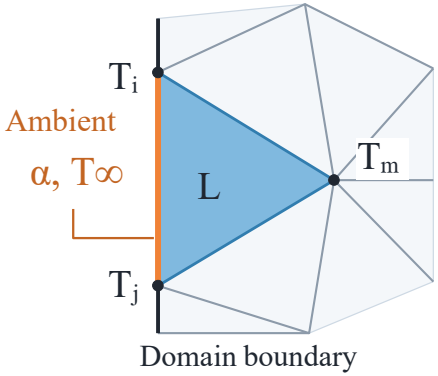
5. Prescribed-flux contribution after integration

$$\Phi_{\Gamma,e}^{(q)} = \frac{\pi \bar{q}_n L}{3} \left[(2r_i + r_j) T_i + (r_i + 2r_j) T_j \right]$$

This linear contribution depends only on the temperatures at boundary nodes i, j.
The positive sign corresponds to heat flux directed outward.

Derivation of the functional: convection

1. Interpolation on edge i, j



L: edge length, $0 \le \xi \le 1$.

$$N_i = 1 - \xi, \quad N_j = \xi$$

$$dL = L\, d\xi$$

$$T^e = N_i T_i + N_j T_j$$

$$r = N_i r_i + N_j r_j$$

2. Convection contribution

$$\alpha = \text{const}, \qquad T_\infty = \text{const}$$

$$\Phi_{\Gamma,e}^{(\alpha)} = \pi\alpha \int_{\Gamma_e} r(T^e - T_\infty)^2\, dL$$

$$= \pi\alpha \int_{\Gamma_e} r(N_i T_i + N_j T_j - T_\infty)^2\, dL$$

3. Expansion of the squared temperature difference

$$(N_i T_i + N_j T_j - T_\infty)^2 = N_i^2 T_i^2 + 2N_i N_j T_i T_j + N_j^2 T_j^2 - 2N_i T_i T_\infty - 2N_j T_j T_\infty + T_\infty^2.$$

4. Nodal temperatures outside the integrals

$$\Phi_{\Gamma,e}^{(\alpha)} = \pi\alpha \Big[T_i^2 \int_{\Gamma_e} r N_i^2\, dL + 2T_i T_j \int_{\Gamma_e} r N_i N_j\, dL + T_j^2 \int_{\Gamma_e} r N_j^2\, dL$$

$$- 2T_i T_\infty \int_{\Gamma_e} r N_i\, dL - 2T_j T_\infty \int_{\Gamma_e} r N_j\, dL + T_\infty^2 \int_{\Gamma_e} r\, dL \Big].$$

The convection contribution contains quadratic, linear, and constant terms.

Convection: evaluation of the edge integrals

1. Notation for geometric integrals

$$I_{ab} = \int_{\Gamma_e} r N_a N_b \, dL, \qquad J_a = \int_{\Gamma_e} r N_a \, dL, \qquad a, b \in \{i, j\}$$

2. Quadratic integral

$$I_{ii} = L \int_0^1 [r_i(1-\xi) + r_j \xi](1-\xi)^2 \, d\xi$$

$$= L\left(\frac{r_i}{4} + \frac{r_j}{12}\right) = \frac{L}{12}(3r_i + r_j).$$

3. Linear integral

$$J_i = L \int_0^1 [r_i(1-\xi) + r_j \xi](1-\xi) \, d\xi$$

$$= L\left(\frac{r_i}{3} + \frac{r_j}{6}\right) = \frac{L}{6}(2r_i + r_j).$$

4. Remaining integrals

$$I_{ij} = I_{ji} = \tfrac{L}{12}(r_i + r_j), \quad I_{jj} = \tfrac{L}{12}(r_i + 3r_j), \quad J_j = \tfrac{L}{6}(r_i + 2r_j), \quad \int_{\Gamma_e} r \, dL = \tfrac{L}{2}(r_i + r_j)$$

5. Scalar functional of the edge

$$\Phi_{\Gamma,e}^{(\alpha)} = \pi\alpha\left[I_{ii}T_i^2 + 2I_{ij}T_iT_j + I_{jj}T_j^2 - 2T_\infty(J_iT_i + J_jT_j) + T_\infty^2 \int_{\Gamma_e} r \, dL\right].$$

$$\Phi_{\Gamma,e}^{(\alpha)} = \frac{\pi\alpha L}{12}\big[(3r_i + r_j)T_i^2 + 2(r_i + r_j)T_iT_j + (r_i + 3r_j)T_j^2$$

$$- 4T_\infty\big((2r_i + r_j)T_i + (r_i + 2r_j)T_j\big) + 6(r_i + r_j)T_\infty^2\big].$$

The integrals are exact for a straight edge with constant α and T∞.

Assembled functional before minimization

1. Complete scalar element functional

$$\Phi_e(\mathbf{T}_e) = \frac{\pi\lambda A_e(r_i + r_j + r_m)}{3D_e^2}\left[(b_iT_i + b_jT_j + b_mT_m)^2 + (c_iT_i + c_jT_j + c_mT_m)^2\right]$$

$$- \frac{\pi Q A_e}{6}\left[(2r_i + r_j + r_m)T_i + (r_i + 2r_j + r_m)T_j + (r_i + r_j + 2r_m)T_m\right]$$

$$+ \frac{\pi}{3}\sum_{\gamma\in\mathcal{E}_{2,e}} \bar{q}_{n,\gamma}L_\gamma\left[(2r_p + r_s)T_p + (r_p + 2r_s)T_s\right]$$

$$+ \frac{\pi}{12}\sum_{\gamma\in\mathcal{E}_{3,e}} \alpha_\gamma L_\gamma\Big[(3r_p + r_s)T_p^2 + 2(r_p + r_s)T_pT_s + (r_p + 3r_s)T_s^2$$

$$- 4T_{\infty,\gamma}\big((2r_p + r_s)T_p + (r_p + 2r_s)T_s\big) + 6(r_p + r_s)T_{\infty,\gamma}^2\Big].$$

$\gamma = (p, s)$: boundary edge; $\mathcal{E}_{2,e},\ \mathcal{E}_{3,e}$: sets of edges with Type II and Type III conditions.

2. Global functional

$$\Phi(\mathbf{T}) = \textstyle\sum_e \Phi_e(\mathbf{T}_e), \qquad \mathbf{T}_e = (T_i, T_j, T_m)^\mathsf{T}$$

All integrals are evaluated. Only nodal temperatures remain unknown. Minimization follows next.

Specialized functional for the piston

1. Assumptions of the teaching model

$$Q = 0, \qquad \bar{q}_n = 0 \quad \Longrightarrow \quad \Phi_e^{(Q)} = \Phi_e^{(q)} = 0$$

2. General functional and vanishing contributions

$$\Phi_e(\mathbf{T}_e) = \frac{\pi\lambda A_e(r_i + r_j + r_m)}{3D_e^2}\left[(b_iT_i + b_jT_j + b_mT_m)^2 + (c_iT_i + c_jT_j + c_mT_m)^2\right]$$

$$- \frac{\pi Q A_e}{6}\left[(2r_i + r_j + r_m)T_i + (r_i + 2r_j + r_m)T_j + (r_i + r_j + 2r_m)T_m\right] \qquad Q = 0$$

$$+ \frac{\pi}{3}\sum_{\gamma\in\mathcal{E}_{2,e}} \bar{q}_{n,\gamma}L_\gamma\left[(2r_p + r_s)T_p + (r_p + 2r_s)T_s\right] \qquad \bar{q}_{n,\gamma} = 0$$

$$+ \frac{\pi}{12}\sum_{\gamma\in\mathcal{E}_{3,e}} \alpha_\gamma L_\gamma\Big[(3r_p + r_s)T_p^2 + 2(r_p + r_s)T_pT_s + (r_p + 3r_s)T_s^2$$

$$- 4T_{\infty,\gamma}\big((2r_p + r_s)T_p + (r_p + 2r_s)T_s\big) + 6(r_p + r_s)T_{\infty,\gamma}^2\Big].$$

γ = (p, s): boundary edge; $\mathcal{E}_{2,e}$, $\mathcal{E}_{3,e}$: sets of edges with Type II and Type III conditions.

Without a Type III edge, convection is absent, but conduction remains.

Next, we minimize the general four-term functional.

Functional minimization

1. Minimum of the assembled functional

For each unknown global nodal temperature:

$$\Phi(\mathbf{T}) = \sum_e \Phi_e(\mathbf{T}_e), \qquad \frac{\partial \Phi}{\partial T_A} = 0$$

2. Nodal equation sums adjacent element contributions

$$\frac{\partial \Phi}{\partial T_A} = \sum_{e:\, A \in e} \frac{\partial \Phi_e}{\partial T_{a(A)}} = 0$$

a(A): local index of global node A. A shared node has one temperature.

3. The derivative of a sum is the sum of derivatives

$$\frac{\partial \Phi_e}{\partial T_a} = \frac{\partial \Phi_e^{(\lambda)}}{\partial T_a} + \frac{\partial \Phi_e^{(Q)}}{\partial T_a} + \frac{\partial \Phi_e^{(q)}}{\partial T_a} + \frac{\partial \Phi_e^{(\alpha)}}{\partial T_a}$$

4. The calculation order is unchanged

$$\lambda \longrightarrow Q \longrightarrow q \longrightarrow \alpha$$

The integrals are evaluated. Nodal polynomial derivatives give matrices and load vectors.

The assembled sum vanishes, not the derivatives of individual elements or contributions.

Minimization: derivative of the conduction functional

1. Scalar element functional

$$\Phi_e^{(\lambda)} = \frac{\pi\lambda A_e(r_i + r_j + r_m)}{3D_e^2}\left[(b_iT_i + b_jT_j + b_mT_m)^2 + (c_iT_i + c_jT_j + c_mT_m)^2\right]$$

2. Derivative of a square

The temperatures are independent. Geometry and λ are constant. a ∈ {i, j, m}.

$$\frac{\partial}{\partial T_a}(b_iT_i + b_jT_j + b_mT_m)^2 = 2b_a(b_iT_i + b_jT_j + b_mT_m)$$

$$\frac{\partial}{\partial T_a}(c_iT_i + c_jT_j + c_mT_m)^2 = 2c_a(c_iT_i + c_jT_j + c_mT_m)$$

3. Functional derivative with the complete coefficient

$$\frac{\partial\Phi_e^{(\lambda)}}{\partial T_a} = \frac{2\pi\lambda A_e(r_i + r_j + r_m)}{3D_e^2}\left[b_a(b_iT_i + b_jT_j + b_mT_m) + c_a(c_iT_i + c_jT_j + c_mT_m)\right]$$

The quadratic functional gives a linear expression in nodal temperatures.

Minimization: element conduction matrix

1. Expanded sums and simplified coefficient

$$D_e^2 = 4A_e^2$$

$$\frac{\partial \Phi_e^{(\lambda)}}{\partial T_a} = \frac{\pi\lambda(r_i + r_j + r_m)}{6A_e}\Big[(b_a b_i + c_a c_i)T_i + (b_a b_j + c_a c_j)T_j + (b_a b_m + c_a c_m)T_m\Big].$$

2. Conduction matrix coefficients

$$K_{e,ab}^{(\lambda)} = \frac{\pi\lambda(r_i + r_j + r_m)}{6A_e}(b_a b_b + c_a c_b), \qquad \frac{\partial \Phi_e^{(\lambda)}}{\partial T_a} = \sum\nolimits_b K_{e,ab}^{(\lambda)} T_b$$

Index a denotes the row: the temperature used for differentiation.
Index b denotes the column: the temperature multiplying the coefficient.

3. Three derivatives form the matrix

$$\mathbf{K}_e^{(\lambda)} = \frac{\pi\lambda(r_i + r_j + r_m)}{6A_e}\begin{pmatrix} b_i^2 + c_i^2 & b_i b_j + c_i c_j & b_i b_m + c_i c_m \\ b_j b_i + c_j c_i & b_j^2 + c_j^2 & b_j b_m + c_j c_m \\ b_m b_i + c_m c_i & b_m b_j + c_m c_j & b_m^2 + c_m^2 \end{pmatrix}$$

Conduction contribution: $\boldsymbol{\partial \Phi_e^{(\lambda)} / \partial \mathbf{T}_e = \mathbf{K}_e^{(\lambda)} \mathbf{T}_e}$

Minimization: volumetric heat source

1. Previously derived functional

$$\Phi_e^{(Q)} = -\frac{\pi Q A_e}{6}\left[(2r_i + r_j + r_m)T_i + (r_i + 2r_j + r_m)T_j + (r_i + r_j + 2r_m)T_m\right]$$

2. Partial derivatives with respect to nodal temperatures

Each temperature enters linearly. Geometry and Q are constant.

$$\frac{\partial \Phi_e^{(Q)}}{\partial T_i} = -\frac{\pi Q A_e}{6}(2r_i + r_j + r_m) \quad \frac{\partial \Phi_e^{(Q)}}{\partial T_j} = -\frac{\pi Q A_e}{6}(r_i + 2r_j + r_m) \quad \frac{\partial \Phi_e^{(Q)}}{\partial T_m} = -\frac{\pi Q A_e}{6}(r_i + r_j + 2r_m)$$

3. Load vector and matrix form

$$\mathbf{F}_e^{(Q)} = \frac{\pi Q A_e}{6}\begin{pmatrix} 2r_i + r_j + r_m \\ r_i + 2r_j + r_m \\ r_i + r_j + 2r_m \end{pmatrix}$$

$$\frac{\partial \Phi_e^{(Q)}}{\partial \mathbf{T}_e} = -\mathbf{F}_e^{(Q)}$$

$$\Phi_e^{(Q)} = -\mathbf{T}_e^{\mathsf{T}}\mathbf{F}_e^{(Q)}$$

For $Q > 0$ and $r \geq 0$, the load is positive in a nondegenerate element.

The source contributes only to the known right-hand side, not to the coefficient matrix.

Minimization: prescribed heat flux

1. Original boundary-edge functional

$$\Phi_{\Gamma,e}^{(q)} = \frac{\pi \bar{q}_n L}{3}\left[(2r_i + r_j)T_i + (r_i + 2r_j)T_j\right]$$

2. Derivatives with respect to nodal temperatures

$$\frac{\partial \Phi_{\Gamma,e}^{(q)}}{\partial T_i} = \frac{\pi \bar{q}_n L}{3}(2r_i + r_j), \qquad \frac{\partial \Phi_{\Gamma,e}^{(q)}}{\partial T_j} = \frac{\pi \bar{q}_n L}{3}(r_i + 2r_j)$$

3. Load vector and heat-flux sign convention

$$\mathbf{F}_{\Gamma,e}^{(q)} = -\frac{\pi \bar{q}_n L}{3}\begin{pmatrix} 2r_i + r_j \\ r_i + 2r_j \end{pmatrix}$$

4. Vector form of the derivative and functional

$$\frac{\partial \Phi_{\Gamma,e}^{(q)}}{\partial \mathbf{T}_\Gamma} = -\mathbf{F}_{\Gamma,e}^{(q)}, \qquad \Phi_{\Gamma,e}^{(q)} = -\mathbf{T}_\Gamma^{\mathsf{T}}\mathbf{F}_{\Gamma,e}^{(q)}, \qquad \mathbf{T}_\Gamma = \begin{pmatrix} T_i \\ T_j \end{pmatrix}$$

Positive outward heat flux gives a negative load.
The linear term does not contribute to the coefficient matrix.

Minimization: derivatives of the convection contribution

1. Original scalar contribution for boundary edge ij

$$\Phi^{(\alpha)} = \frac{\pi\alpha L}{12}\Big[(3r_i + r_j)T_i^2 + 2(r_i + r_j)T_iT_j + (r_i + 3r_j)T_j^2 - 4T_\infty\big((2r_i + r_j)T_i + (r_i + 2r_j)T_j\big) + 6(r_i + r_j)T_\infty^2\Big].$$

α, L, r_i, r_j, and T∞ are given. Only nodal temperatures T_i and T_j are unknown.

2. Derivative with respect to T_i

$$\frac{\partial\Phi^{(\alpha)}}{\partial T_i} = \frac{\pi\alpha L}{12}\Big[2(3r_i + r_j)T_i + 2(r_i + r_j)T_j - 4T_\infty(2r_i + r_j)\Big] = \frac{\pi\alpha L}{6}\Big[(3r_i + r_j)T_i + (r_i + r_j)T_j - 2T_\infty(2r_i + r_j)\Big].$$

3. Derivative with respect to T_j

$$\frac{\partial\Phi^{(\alpha)}}{\partial T_j} = \frac{\pi\alpha L}{12}\Big[2(r_i + r_j)T_i + 2(r_i + 3r_j)T_j - 4T_\infty(r_i + 2r_j)\Big] = \frac{\pi\alpha L}{6}\Big[(r_i + r_j)T_i + (r_i + 3r_j)T_j - 2T_\infty(r_i + 2r_j)\Big].$$

Constant term: $\boldsymbol{\partial\big[(\pi\alpha L/2)(r_i + r_j)T_\infty^2\big]/\partial T_a = 0, \quad a \in \{i, j\}.}$

Minimization: convection matrix and load

1. The two derivatives as a vector

$$\mathbf{T}_\Gamma = \begin{pmatrix} T_i \\ T_j \end{pmatrix}$$

$$\frac{\partial \Phi^{(\alpha)}}{\partial \mathbf{T}_\Gamma} = \frac{\pi \alpha L}{6} \begin{pmatrix} 3r_i + r_j & r_i + r_j \\ r_i + r_j & r_i + 3r_j \end{pmatrix} \mathbf{T}_\Gamma - \frac{\pi \alpha T_\infty L}{3} \begin{pmatrix} 2r_i + r_j \\ r_i + 2r_j \end{pmatrix}.$$

The first row is the derivative with respect to T_i; the second is with respect to T_j.

2. Convection matrix and load vector

$$\mathbf{K}^{(\alpha)} = \frac{\pi \alpha L}{6} \begin{pmatrix} 3r_i + r_j & r_i + r_j \\ r_i + r_j & r_i + 3r_j \end{pmatrix}$$

$$\mathbf{F}^{(\alpha)} = \frac{\pi \alpha T_\infty L}{3} \begin{pmatrix} 2r_i + r_j \\ r_i + 2r_j \end{pmatrix}$$

3. Matrix and right-hand-side contributions

The flux term αT depends on the unknown surface temperature and gives the matrix.

The known term αT∞ gives the load. Together they represent the Type III (Robin) condition.

Convection contribution: $\boldsymbol{\partial \Phi^{(\alpha)} / \partial \mathbf{T}_\Gamma = \mathbf{K}^{(\alpha)} \mathbf{T}_\Gamma - \mathbf{F}^{(\alpha)}}$

Minimization: combined element derivatives

1. Sum of all differentiated contributions

$$\mathbf{R}_e = \frac{\partial \Phi_e}{\partial \mathbf{T}_e} = \left(\mathbf{K}_e^{(\lambda)} + \mathbf{K}_e^{(\alpha)}\right)\mathbf{T}_e - \left(\mathbf{F}_e^{(Q)} + \mathbf{F}_e^{(q)} + \mathbf{F}_e^{(\alpha)}\right) = \mathbf{K}_e\mathbf{T}_e - \mathbf{F}_e$$

2. Boundary matrix in the edge-node positions

$$\mathbf{K}_{ij}^{(\alpha)} = \frac{\pi\alpha L_{ij}}{6}\begin{pmatrix} 3r_i + r_j & r_i + r_j \\ r_i + r_j & r_i + 3r_j \end{pmatrix} \quad \longrightarrow \quad \mathbf{K}_e^{(\alpha)} = \frac{\pi\alpha L_{ij}}{6}\begin{pmatrix} 3r_i + r_j & r_i + r_j & 0 \\ r_i + r_j & r_i + 3r_j & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Row and column order: i, j, m. Edge ij contributes zero at node m.

3. Example for the next slide

Edge ij: convection to the surroundings. Edge jm: prescribed outward heat flux.
The element contains source Q. Edge im gives no additional contribution.

The assembled sum vanishes, not the residual R_e of each element separately.

General example: expanded matrices

1. Five contributions to the element functional derivative

Type III on ij; outward flux on jm; volumetric source Q; no contribution on im.

$$\frac{\partial \Phi_e}{\partial \mathbf{T}_e} = \begin{pmatrix} R_{e,i} \\ R_{e,j} \\ R_{e,m} \end{pmatrix} = \frac{\pi\lambda(r_i + r_j + r_m)}{6A_e} \begin{pmatrix} b_i^2 + c_i^2 & b_i b_j + c_i c_j & b_i b_m + c_i c_m \\ b_j b_i + c_j c_i & b_j^2 + c_j^2 & b_j b_m + c_j c_m \\ b_m b_i + c_m c_i & b_m b_j + c_m c_j & b_m^2 + c_m^2 \end{pmatrix} \begin{pmatrix} T_i \\ T_j \\ T_m \end{pmatrix}$$

$$+ \frac{\pi\alpha L_{ij}}{6} \begin{pmatrix} 3r_i + r_j & r_i + r_j & 0 \\ r_i + r_j & r_i + 3r_j & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} T_i \\ T_j \\ T_m \end{pmatrix}$$

$$- \frac{\pi Q A_e}{6} \begin{pmatrix} 2r_i + r_j + r_m \\ r_i + 2r_j + r_m \\ r_i + r_j + 2r_m \end{pmatrix} + \frac{\pi \bar{q}_n L_{jm}}{3} \begin{pmatrix} 0 \\ 2r_j + r_m \\ r_j + 2r_m \end{pmatrix} - \frac{\pi\alpha T_\infty L_{ij}}{3} \begin{pmatrix} 2r_i + r_j \\ r_i + 2r_j \\ 0 \end{pmatrix}.$$

2. The same contribution in compact matrix form

$$\frac{\partial \Phi_e}{\partial \mathbf{T}_e} = \begin{pmatrix} R_{e,i} \\ R_{e,j} \\ R_{e,m} \end{pmatrix} = \begin{pmatrix} K_{e,ii} & K_{e,ij} & K_{e,im} \\ K_{e,ji} & K_{e,jj} & K_{e,jm} \\ K_{e,mi} & K_{e,mj} & K_{e,mm} \end{pmatrix} \begin{pmatrix} T_i \\ T_j \\ T_m \end{pmatrix} - \begin{pmatrix} F_{e,i} \\ F_{e,j} \\ F_{e,m} \end{pmatrix}$$

The assembled global residual vanishes, not the residual of an individual element.

§ 21. Piston special case: three contributions remain

1. Zero source and prescribed heat flux

$$Q = 0, \quad \bar{q}_n = 0 \quad \Longrightarrow \quad \mathbf{F}_e^{(Q)} = \mathbf{F}_e^{(q)} = \mathbf{0}$$

2. Expanded derivative: convection on edge ij

$$\frac{\partial \Phi_e}{\partial \mathbf{T}_e} = \begin{pmatrix} R_{e,i} \\ R_{e,j} \\ R_{e,m} \end{pmatrix} = \frac{\pi\lambda(r_i + r_j + r_m)}{6A_e} \begin{pmatrix} b_i^2 + c_i^2 & b_i b_j + c_i c_j & b_i b_m + c_i c_m \\ b_j b_i + c_j c_i & b_j^2 + c_j^2 & b_j b_m + c_j c_m \\ b_m b_i + c_m c_i & b_m b_j + c_m c_j & b_m^2 + c_m^2 \end{pmatrix} \begin{pmatrix} T_i \\ T_j \\ T_m \end{pmatrix}$$

$$+ \frac{\pi\alpha L_{ij}}{6} \begin{pmatrix} 3r_i + r_j & r_i + r_j & 0 \\ r_i + r_j & r_i + 3r_j & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} T_i \\ T_j \\ T_m \end{pmatrix} - \frac{\pi\alpha T_\infty L_{ij}}{3} \begin{pmatrix} 2r_i + r_j \\ r_i + 2r_j \\ 0 \end{pmatrix}.$$

3. Compact element form

$$\mathbf{K}_e^{\mathrm{p}} = \mathbf{K}_e^{(\lambda)} + \mathbf{K}_e^{(\alpha)}$$

$$\frac{\partial \Phi_e}{\partial \mathbf{T}_e} = \begin{pmatrix} R_{e,i} \\ R_{e,j} \\ R_{e,m} \end{pmatrix} = \begin{pmatrix} K_{e,ii}^{\mathrm{p}} & K_{e,ij}^{\mathrm{p}} & K_{e,im}^{\mathrm{p}} \\ K_{e,ji}^{\mathrm{p}} & K_{e,jj}^{\mathrm{p}} & K_{e,jm}^{\mathrm{p}} \\ K_{e,mi}^{\mathrm{p}} & K_{e,mj}^{\mathrm{p}} & K_{e,mm}^{\mathrm{p}} \end{pmatrix} \begin{pmatrix} T_i \\ T_j \\ T_m \end{pmatrix} - \begin{pmatrix} F_{e,i}^{(\alpha)} \\ F_{e,j}^{(\alpha)} \\ F_{e,m}^{(\alpha)} \end{pmatrix}$$

For several Type III edges, sum their matrices and loads in the corresponding positions.

Only the sum of element contributions vanishes after global assembly.

Global assembly: nodal temperature derivatives

1. Global functional

$$\Phi(\mathbf{T}) = \sum_{e=1}^{n_e} \Phi_e, \qquad \underbrace{\mathbf{T} = (T_1 \ \cdots \ T_i \ \cdots \ T_j \ \cdots \ T_m \ \cdots \ T_N)^{\mathsf{T}}}_{N\times 1}$$

N: number of mesh nodes; n_e: number of elements. All nodal temperatures are unknown.

2. Derivatives with respect to all N temperatures

$$\underbrace{\frac{\partial \Phi}{\partial \mathbf{T}}}_{N\times 1} = \begin{pmatrix} \frac{\partial \Phi}{\partial T_1} \\ \vdots \\ \frac{\partial \Phi}{\partial T_N} \end{pmatrix} = \sum_{e=1}^{n_e} \begin{pmatrix} \frac{\partial \Phi_e}{\partial T_1} \\ \vdots \\ \frac{\partial \Phi_e}{\partial T_N} \end{pmatrix} = \mathbf{0}$$

3. Contributions only at the element nodes

$$\frac{\partial \Phi_e}{\partial T_n} = \begin{cases} R_{e,n}, & n \in \{i,j,m\}, \\ 0, & n \notin \{i,j,m\}. \end{cases}$$

Contributions at i, j, m may be nonzero.
A shared node has one temperature in all adjacent elements.

The total derivative vanishes: $\boldsymbol{\partial \Phi / \partial \mathbf{T} = 0}$

Global assembly: matrix and load vector

1. Element contributions mapped to global node numbers

$$\underbrace{\frac{\partial \Phi}{\partial \mathbf{T}}}_{N\times 1} = \Bigg[\underbrace{\begin{pmatrix} K_{11}^{(1)} & K_{12}^{(1)} & K_{13}^{(1)} & 0 & \cdots & 0 \\ K_{21}^{(1)} & K_{22}^{(1)} & K_{23}^{(1)} & 0 & \cdots & 0 \\ K_{31}^{(1)} & K_{32}^{(1)} & K_{33}^{(1)} & 0 & \cdots & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 \end{pmatrix}_{N\times N}}_{\widetilde{\mathbf{K}}^{(1)}} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & K_{22}^{(2)} & K_{23}^{(2)} & K_{24}^{(2)} & \cdots & 0 \\ 0 & K_{32}^{(2)} & K_{33}^{(2)} & K_{34}^{(2)} & \cdots & 0 \\ 0 & K_{42}^{(2)} & K_{43}^{(2)} & K_{44}^{(2)} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 \end{pmatrix}_{N\times N}}_{\widetilde{\mathbf{K}}^{(2)}} + \cdots \Bigg] \begin{pmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \\ \vdots \\ T_N \end{pmatrix}$$

$$- \Bigg[\underbrace{\begin{pmatrix} F_1^{(1)} \\ F_2^{(1)} \\ F_3^{(1)} \\ 0 \\ \vdots \\ 0 \end{pmatrix}_{N\times 1}}_{\widetilde{\mathbf{F}}^{(1)}} + \underbrace{\begin{pmatrix} 0 \\ F_2^{(2)} \\ F_3^{(2)} \\ F_4^{(2)} \\ \vdots \\ 0 \end{pmatrix}_{N\times 1}}_{\widetilde{\mathbf{F}}^{(2)}} + \cdots \Bigg] = \mathbf{0}.$$

At shared nodes, add contributions at matching global positions.

$$\frac{\partial \Phi}{\partial \mathbf{T}} = \underbrace{\left(\widetilde{\mathbf{K}}^{(1)} + \widetilde{\mathbf{K}}^{(2)} + \cdots + \widetilde{\mathbf{K}}^{(n_e)}\right)}_{\mathbf{K}} \mathbf{T} - \underbrace{\left(\widetilde{\mathbf{F}}^{(1)} + \widetilde{\mathbf{F}}^{(2)} + \cdots + \widetilde{\mathbf{F}}^{(n_e)}\right)}_{\mathbf{F}} = \mathbf{0}.$$

Global system:

$$\underbrace{\mathbf{K}}_{N\times N}\ \underbrace{\mathbf{T}}_{N\times 1} = \underbrace{\mathbf{F}}_{N\times 1}$$

Solution and temperature field recovery

1. Mesh nodal temperatures

$$\mathbf{KT} = \mathbf{F}$$

Solving the global system gives the temperatures at every mesh node.

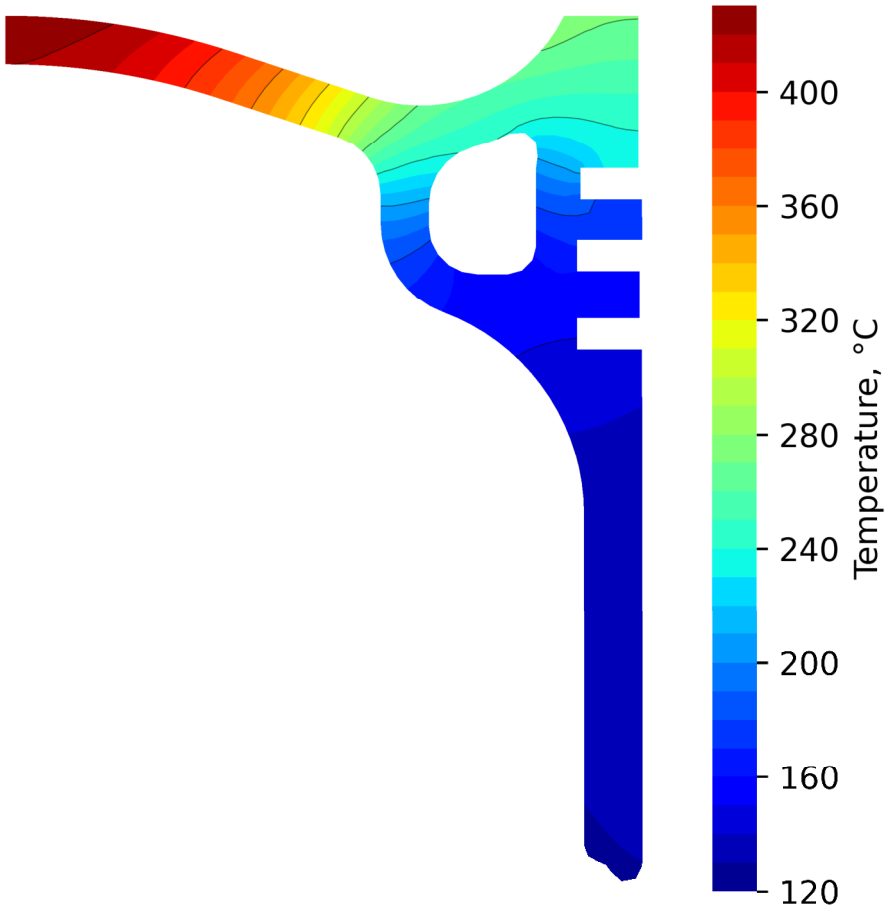
2. Temperature within an element

$$T^e(r,z) = N_i(r,z)T_i + N_j(r,z)T_j + N_m(r,z)T_m$$

Three nodal temperatures and shape functions define the distribution within the triangle.

3. Temperature field of the whole component

Interpolation over all elements gives an approximate temperature field of the component.



Piston temperature field

Nodal temperatures → element fields → temperature field of the whole component



Bauman Moscow State Technical University

Special Topics in Finite Element Modeling for IC engines

CONCLUSION